

YINYUAN ZHAO

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EDUCATION

University of California, Santa Cruz | California, USA

Jun. 2023 – Present

- *PhD in Computer Science and Engineering*
- Focus: x86 CPU Micro-Architecture, Operating System

South China University of Technology | Guangdong, China

Sept. 2016 – Jun. 2020

- *Bachelor of Software Engineering*
- Focus: Chiplet SoC, Embedded System

PUBLICATIONS

- **Y. Zhao**, S. Oh, M. Xu, and H. Litz, "ATR: Out-of-Order Register Release Exploiting Atomic Regions," (MICRO) *International Symposium on Microarchitecture*, 2025. **TOP-TIER CONFERENCE**
- **Y. Zhao**, X. Wang, Y. Jiang, L. Wang, A.K. Singh, L. Huang, and M. Yang, "An Enhanced Planned Obsolescence Attack by Aging Networks-on-Chip," (JSA) *Journal of Systems Architecture*, 2021.
- S.S. Sahoo, A. Kumar, M. Decky, S.C.B. Wong, G.V. Merrett, **Y. Zhao**, J. Wang, X. Wang, and A.K. Singh, "Emergent Design Challenges for Embedded Systems and Paths Forward: Mixed-criticality, Energy, Reliability and Security Perspectives," (CODES+ISSS) *International Conference on Hardware/Software Codesign and System Synthesis*, 2021.
- **Y. Zhao** and X. Wang, "A Hardware Aging Attack Method for a Network-on-Chip," *CN Patent*, 2019.

WORK EXPERIENCE

CPU Performance Engineer Intern | *Qualcomm* | Santa Clara, California, USA

Jun. 2026 – Sept. 2026

- **Profiling Infrastructure**: Developed an automated profiling infrastructure using Linux Perf and ARM Top-down tool to collect PMU events and calculate performance metrics across diverse platforms.
- **CPU Performance Analysis**: Conducted PMU-based performance analysis to identify performance bottlenecks in critical workloads, delivering optimization insights to improve CPU performance.

Full-Time Embedded Engineer | *Digital Power* | Shanghai, China

Mar. 2021 – Sept. 2022

- **Linux Kernel Device**: Designed a multiprocessor shared-memory driver by developing multi-layered in-kernel message queues and DTS configurations, supporting multi-thread asynchronous communication.
- **RTOS Driver**: Built an IoT edge system on RTOS by implementing communication device drivers for UART, CAN, Ethernet, I2C, SPI, and BLE, integrating TCP/IP protocols and sockets.
- **Trusted Firmware**: Developed a boot loader based on U-Boot for a new ARMv8-based chip, achieving device initialization and authentication across BL2-BL33 stages leveraging TF-A.

RESEARCH PROJECTS

CPU Core Micro-Architecture [[GitHub](#)]

University of California, Santa Cruz | California, USA

- Modeled key components include register, issue queue, and LSQ in a C++ processor simulator.
- Implemented a top-down performance analysis method enabling hierarchical bottleneck decomposition.
- Architected a novel register renaming scheme by register early releasing, improving SPEC2017 IPC by 5.13%.

Network-on-Chip Chiplet

South China University of Technology | Guangdong, China

- Designed an aging attack method targeting chip lifetime degradation based on two proposed routing algorithms in multiprocessor SoCs, simulated in C and Verilog, and evaluated through Synopsys DC synthesis.